Personal Protection Equipment.

Introduction

There are so many potential hazards at a construction site, and personal protective equipment (PPE) is one of the workers' primary lines of defense. Each item is designed to minimize exposure to certain risks of injury or illness, as well as the damage caused by those hazards in the event of a mishap**.**

The Importance of Personal Protective Equipment

Protecting you from Hazards; Personal protective equipment is NOT the first line of defense against identified hazards at our workplace. We make every attempt to engineer solutions to prevent injuries so PPE isn't necessary. When we can't fully eliminate or control the hazard, then we provide PPE for your safety.

When and where is Protection Required?

The company assesses all the work areas to find the hazards that are present, or might be present at some time, and then makes protective equipment decisions based on those assessments. This is an ongoing process to make sure we are always current and aware of the hazards of our work environment. Supervisors will keep you informed of the type of PPE for your work area or the specific jobs you perform.

Proper Use

No PPE will do you any good unless it is sized correctly and fitting properly. This may sound elementary, but everything from gloves to shoes to glasses to goggles must fit the right way in order for it to properly protect you. This is why you need to take some time to make sure all of your PPE is the correct size. If some equipment doesn't fit in the right way, wearing it is not doing you any good at all.

PPE has its Limitations

PPE is effective when it is used as designed. For instance, not all gloves protect against all hazards: leather gloves provide good protection against cuts and slivers but do nothing for chemical safety.

Safety glasses protect your eyes but not the rest of your face. For each type of PPE, it is important to know what it will guard against and what it won't

Proper Care and Maintenance

PPE that is not properly maintained will not last long and will not protect you. In fact, poorly maintained equipment can be a greater hazard: glasses that are dirty create visibility problems, chemical gloves with pin holes will allow chemicals to touch your skin and shoes held together by tape won't protect your feet and will create trip hazards.

Before each use, inspect equipment for potential problems, and never use damaged PPE. Contact your supervisor and only proceed with the task when you are equipped with properly maintained PPE. When you are done using the gear, clean it up so it is ready for the next use.

The different types of PPE include:

- ✓ Head and scalp protection;
- ✓ Respiratory protection;
- ✓ Eye protection;
- ✓ Hearing protection;
- ✓ Hand and arm protection;
- ✓ Foot and leg protection;
- ✓ Body protection;
- ✓ Height and access protection

The Requirement for PPE

To ensure the greatest possible protection for employees in the workplace, the cooperative efforts of both employers and employees will help in establishing and maintaining a safe and healthful work environment. In general, employers are responsible for:

- Performing a "hazard assessment" of the workplace to identify and control physical and health hazards.
- Identifying and providing appropriate PPE for employees.
- Training employees in the use and care of the PPE.
- Maintaining PPE, including replacing worn or damaged PPE.
- Periodically reviewing, updating and evaluating the effective- ness of the PPE program. In general, employees should:
- Properly wear PPE
- Attend training sessions on PPE
- Care for, clean and maintain PPE, and
- Inform a supervisor of the need to repair or replace PPE.

PPE Assessment

In order to determine what types of PPE are necessary, supervisors must first identify physical and health hazards in the workplace. This process is known as conducting a PPE Assessment. The PPE Assessment process must be documented using the PPE Assessment form (PPE Program, Appendix A). PPE Assessments should include the following:

• A walkthrough of the workplace to develop a list of potential hazards

• Identification of all physical, biological, chemical, and radioactive hazards

• A review of any history of occupational illnesses or injuries

• Identification of the appropriate required PPE for the hazard/task RESPSONSIBILITIES Supervisors:

• Conduct a PPE Assessment for each job position or title under their direction

• Ensure that PPE is provided to employees and that it is used properly Employees:

• Must utilize the provided PPE in accordance with the instruction and training provided

• Employees must inspect their PPE, properly care for and store their PPE, and attend PPE training

SELECTING PPE

• When selecting PPE, fit and comfort should be taken into consideration

• Most PPE comes in a variety of sizes and care should be taken to select the proper size for each employee

• Improperly fitted PPE may not provide the level of protection desired and may discourage employee use

• OSHA and Cornell University require that many categories of PPE meet or be equivalent to standards developed by the American National Standards Institute (ANSI) and the National Institute for Occupational Safety and Health (NIOSH).

Training Employees in the Proper Use of PPE Employers are required to train each employee who must use PPE. Employees must be trained to know at least the following:

- When PPE is necessary.
- What PPE is necessary
- How to properly put on, take off, adjust and wear the PPE.
- The limitations of the PPE.
- Proper care, maintenance, useful life and disposal of PPE.

Employers should make sure that each employee demonstrates an understanding of the PPE training as well as the ability to properly wear and use PPE before they are allowed to perform work requiring the use of the PPE. If an employer believes that a previously trained employee is not demonstrating the proper understanding and skill level in the use of PPE, that employee should receive retraining.

Other situations that require additional or retraining of employees include the following circumstances: changes in the workplace or in the type of required PPE that make prior training obsolete. The employer must document the training of each employee required to wear or use PPE by preparing a certification containing the name of each employee trained, the date of training and a clear identification of the subject of the certification.

Protective Clothing:

At its simplest term protective clothing means overalls or lab coats for general-purpose use. They are intended to protect the user (or the user's own clothing) from every day wear, tear or dirt. There are a number of special hazards that may be encountered against which such basic clothing may not be adequate: Corrosive liquids: could soak into the clothing and so come in contact with the skin, causing serious damages. Impermeability is an important factor here. Dust retention: When working with powders, a fabric that holds dust could generate an airborne exposure hazard as the person moves around. Thermal environment: normal clothing may be too warm or too cold for a particular environment. In extremes cases, chemical protective clothing might be necessary. Typically this comprises a one-piece suit made from an impervious material.

Body protection

The Regulations' definition of PPE excludes ordinary working clothes and uniforms which have no specific protection for the wearer. However, body protection may be required for extended periods of work outdoors to protect against the weather, and to ensure high visibility during work where there is mixed vehicle and pedestrian traffic.

PPE for the body may also be required where workers are exposed to extremes of temperature (whether outdoors or indoors), as well as chemical or metal splash, spray from pressure leaks or spray guns, impact or penetration, contaminated dust, excessive wear, entanglement of own clothing or the risk of drowning.

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When choosing body protection the following factors should be considered:

- Thermal comfort, for example, due to sweating;
- Cost and practicality of cleaning;
- Emergency procedures, such as buoyancy or the need to be identified or spotted in hazardous situations;
- Level of hygiene control required;
- Level of personal contamination;
- Personal preference;
- Restriction of movement;
- Storage;
- Temperature and humidity fluctuation;
- Whether the worker is involved in a process that is wet or dry.

Every construction personnel should wear adequate and suitable protective clothing to protect them from possible hazard due to weather changes, electricity, impacts, and other risks. For Site Engineers a reflected vest is recommended to maintain visibility when doing site inspections and other site activities.

Foot Protection/Safety Shoes

Safety shoes should be worn by all of the construction personnel at the worksite to protect their feet for possible injuries. The safety shoes to be used shall be determined according to potential hazards and according to the nature of work. It shall comply with the safety standard and specification. For worksite safety shoes with steel toe cap covering and steel sole should be considered, although an equivalent material is also acceptable as long as it follows the safety standards. If the personnel is working in electrical installation, non-contractive footwear shall be used.

Examples of situations in which an employee should wear foot and/or leg protection include:

- When heavy objects such as barrels or tools might roll onto or fall on the employee's feet;
- Working with sharp objects such as nails or spikes that could pierce the soles or uppers of ordinary shoes;
- Exposure to molten metal that might splash on feet or legs;
- Working on or around hot, wet or slippery surfaces;
- Working when electrical hazards are present

Foot and leg protection choices include the following:

Leggings protect the lower legs and feet from heat hazards such as molten metal or welding sparks. Safety snaps allow leggings to be removed quickly.

Metatarsal guards protect the instep area from impact and compression. Made of aluminum, steel, fiber or plastic, these guards may be strapped to the outside of shoes.

Toe guards fit over the toes of regular shoes to protect the toes from impact and compression hazards. They may be made of steel, aluminum or plastic. 21 Combination foot and shin guards protect the lower legs and feet, and may be used in combination with toe guards when greater protection is needed.

Safety shoes have impact-resistant toes and heat-resistant soles that protect the feet against hot work surfaces common in roofing, paving and hot metal industries. The metal insoles of some safety shoes protect against puncture wounds. Safety shoes may also be designed to be electrically conductive to prevent the buildup of static electricity in areas with the potential for explosive atmospheres or nonconductive to protect employees from workplace electrical hazards.

1. Head Protection/Safety Helmets/Hard Hat

In construction worksite, there is a higher chance of possible head injury due to falling debris and other related accidents. Site Engineers and all construction personnel involved in site activities should wear a safety helmet or hard hat before entering the work site and it should be wear at all times.

The safety helmets/hard hat shall meet the specifications contained in the technical guidelines issued by the Specialize Department in accordance to international standards- OSHA. Safety helmets/hard hats should be inspected by the safety officer to ensure that it is safe and reliable to use. It should be free from cracks and proper shock-absorbing lining of the helmet should be in good condition.

Employers must ensure that their employees wear head protection if any of the following apply: n Objects might fall from above and strike them on the head; they might bump their heads against fixed objects, such as exposed pipes or beams; or there is a possibility of accidental head contact with electrical hazards. Some examples of occupations in which employees should be required to wear head protection include construction workers, carpenters, electricians, linemen, plumbers and pipefitters, timber and log cutters, welders, among many others.

In general, protective helmets or hard hats should do the following:

- Resist penetration by objects.
- Absorb the shock of a blow.
- Be water-resistant and slow burning.
- Have clear instructions explaining proper adjustment and replacement of the suspension and headband

Types of Hard Hats

Hard hats are divided into three industrial classes:

- **Class A hard hats** provide impact and penetration resistance along with limited voltage protection (up to 2,200 volts).
- **Class B hard hats** provide the highest level of protection against electrical hazards, with high-voltage shock and burn protection (up to 20,000

volts). They also provide protection from impact and penetration hazards by flying/falling objects.

- **Class C hard hats** provide lightweight comfort and impact protection but offer no protection from electrical hazards.

Hard hats with any of the following defects should be removed from service and replaced: n Perforation, cracking, or deformity of the brim or shell; n Indication of exposure of the brim or shell to heat, chemicals or ultraviolet light and other radiation (in addition to a loss of surface gloss, such signs include chalking or flaking).

Eyes and Face Protections

Construction employees shall be provided with eye and face protection equipment when their job had a potential risk of exposing the eyes and face from hazard. They are the machines operators, welder, bar bender and the like. Those workers whose vision requires the use of medical spectacles shall be protected by eyeglasses with protected lenses, goggles that can be worn without disturbing their work activity.

Eyes and face protection equipment shall be free from structural and optical defects. Just like other PPE, it should also pass the required safety standards.

OSHA suggests that eye protection be routinely considered for use by carpenters, electricians, machinists, mechanics, millwrights, plumbers and pipefitters, sheetmetal employees and tinsmiths, assemblers, sanders, grinding machine operators, sawyers, welders, laborers, chemical pro-cess operators and handlers, and timber cutting and logging workers. Employers of employees in other job categories should decide whether there is a need for eye and face PPE through a hazard assessment. Examples of potential eye or face injuries include: Dust, dirt, metal or wood chips entering the eye from activities such as chipping, grinding, sawing, hammering, the use of power tools or even strong wind forces.

Chemical splashes from corrosive substances, hot liquids, solvents or other hazardous solutions.

Objects swinging into the eye or face, such as tree limbs, chains, tools or ropes.

Radiant energy from welding, harmful rays from the use of lasers or other radiant light (as well as heat, glare, sparks, splash and flying particles)

Types of Eye Protection Selecting the most suitable eye and face protection for employees should take into consideration the following elements:

- Ability to protect against specific workplace hazards.
- Should fit properly and be reasonably comfortable to wear.
- Should provide unrestricted vision and movement.
- Should be durable and cleanable.
- Should allow unrestricted functioning of any other required PPE.
- The eye and face protection selected for employee use must clearly identify the manufacturer.

Some of the most common types of eye and face protection include the following:

Safety spectacles. These protective eyeglasses have safety frames constructed of metal or plastic and impact-resistant lenses. Side shields are available on some models.

Goggles. These are tight-fitting eye protection that completely cover the eyes, eye sockets and the facial area immediately surrounding the eyes and provide protection from impact, dust and splashes. Some goggles will fit over corrective lenses.

Welding shields. Constructed of vulcanized fiber or fiberglass and fitted with a filtered lens, welding shields protect eyes from burns caused by infrared or intense radiant light; they also protect both the eyes and face from flying sparks, metal spatter and slag chips produced during welding, brazing, soldering and

Laser safety goggles. These specialty goggles protect against intense concentrations of light produced by lasers. The type of laser safety goggles an employer chooses will depend upon the equipment and operating conditions in the workplace.

Face shields. These transparent sheets of plastic extend from the eyebrows to below the chin and across the entire width of the employee's head. Some are polarized for glare protection. Face shields protect against nuisance dusts and

potential splashes or sprays of hazardous liquids but will not provide adequate protection against impact hazards. Face shields used in combination with goggles or safety spectacles will provide additional protection against impact hazards.

Hearing Protection

Hearing protection must be worn whenever the noise level exceeds the noise exposure standard and the client has implemented hearing protection as a control. You will be told if hearing protection is required. You must wear this protection at all times in designated areas to protect your hearing and to protect you from noise induced hearing loss. As an employee the client must provide you with periodic hearing tests and audiometric testing (if required).

And in the event that noise levels in worksite are too much to handle, ear protected gear are also needed by the site personnel. Ear protected hazard inserted into the ear in the form of a headset, foam earplug and earmuffs shall be in accordance with international safety standards. Plain cotton to be placed in the ear just to block construction noise is prohibited to use.

5. Hand Protection

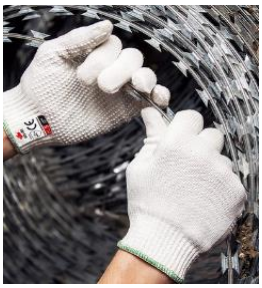
All personnel handling rough, sharp and excessive coarse material such as reinforcement bars, rods, pre-cast concrete, toxic, electric and hot materials shall be protected by safety gloves. The material of these hand protections shall be suitable against the hazard of doing such activities. Site employees shall be protected by the different type of gloves according to their job description. For example, chrome leather gloves should be used when handling sandblasting and other materials. Gloves that are made of fabrics are used for handling rough materials. Insulated rubber gloves should be used by electricians.

The following are examples of some factors that may influence the selection of protective gloves for a workplace.

- Type of chemicals handled.
- Nature of contact (total immersion, splash, etc.).
- Duration of contact.
- Area requiring protection (hand only, forearm, arm).
- Grip requirements (dry, wet, oily).
- Thermal protection.
- Size and comfort. Abrasion/resistance requirements. Gloves made from a wide variety of materials are designed for many types of workplace hazards.

In general, gloves fall into four groups:

- Gloves made of leather, canvas or metal mesh;
- Fabric and coated fabric gloves;
- Chemical- and liquid-resistant gloves;
- Insulating rubber gloves (See 29 CFR 1910.137 and the following section on electrical protective equipment for detailed requirements on the selection, use and care of insulating rubber gloves).



Respiratory Protection:

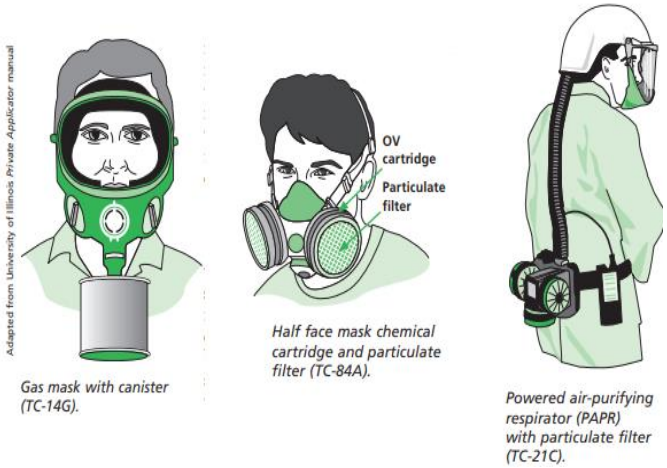
Used to protect the individual wearer against the inhalation of hazardous substances in the workplace air. RPE should only be used where adequate control of exposure cannot be achieved by other means, in other words, as a last resort within the hierarchy of control measures: Elimination, Substitution, Engineering Controls, Administrative Controls, PPE.

Elements of a Respiratory Programme

- ✓ A written plan detailing how the programme is managed
- ✓ A complete assessment and knowledge of respiratory hazards that will be encountered in the workplace
- ✓ Procedures and equipment to control respiratory hazards, including the use of engineering controls and work practices designed to limit or reduce employee exposures to such hazards
- ✓ Guidelines for the proper selection of appropriate respiratory protective equipment
- ✓ An employee training program covering hazard recognition, the dangers associated with respiratory hazards, and proper care and use of respiratory protective equipment
- ✓ Inspection, maintenance, and repair of respiratory protective equipment
- ✓ Medical surveillance of employees, where necessary

Types of Respirators;

Air-purifying respirators (APRs); Air-purifying respirators can remove contaminants in the air that you breathe by filtering out particulates (e.g., dusts, metal fumes, mists, etc.)



Supplied-air respirators (SARs); supply clean air from a compressed air tank or through an airline. This air is not from the work room area. The air supplied in tanks or from compressors must meet certain standards for purity and moisture content.

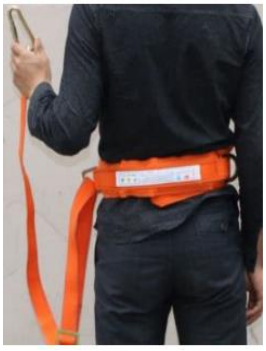
Storing Respirators;

Protect against dust, sunlight, heat, extreme cold excessive moisture and damaging chemicals. Store in position to retain natural configuration.

• Fall Protection: Fall Protection Equipment.

7. Safety Belts/Safety Harness

Site employees working at an elevated level to about 2m or more shall be provided by safety harness or belts to protect them from accidental fall.



Advantages of Air-purifying respirators (APRs)

- Provides the highest level of protection against airborne contamination and oxygen deficiency
- Provides highest level of protection under strenuous work conditions
- Enhance mobility
- Enables longer work periods
- Protects against most airborne contaminations

Disadvantages of Air-purifying respirators (APRs)

- Only protects against specific chemicals and up to specific concentrations
- Limited duration of protection
- May impair movement in confined spaces
- As the length of the hose is increased the minimum approved airflow may not be delivered at the faceplate
- Air-line is vulnerable to damage, chemical contamination, and degradation, decontamination of hoses may be difficult.
- Worker must retrace steps to leave work area.